SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

SUSTAINABLE DEVELOPMENT GOALS (SDG) ASSIGNMENT

SDG 1: No Poverty

Course Code	**CSA1739**	**Course Title**	**Artificial Intelligence**
CO Assessed	**CO3**	**Assessment**	**SDG**
Weightage	**30%**	**Total Marks**	**100**
CO3 Statement	*Implement machine learning algorithms by developing Decision Tree, Random Forest, and Support Vector Machine (SVM) classifiers using the Iris dataset, applying the Gradient Descent algorithm, and performing image segmentation using the K-Means algorithm.*		
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Section/Batch	**Slot D / 2023**	**Date**	**03-11-2026**

Code of Conduct:

I ***B Reddy Balu (192312499),*** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

A Propositional Logic-Based Knowledge System for Identifying Families Eligible for Government Poverty-Alleviation Schemes

1. Introduction

Poverty remains one of the most persistent development challenges. Sustainable Development Goal 1 (No Poverty) aims to end poverty in all its forms everywhere, and one of its key targets is to implement nationally appropriate social protection systems so that the poor and vulnerable are well covered. In practice, governments run many schemes covering housing, food security, pensions, education support, employment guarantee and health insurance. The difficulty is not only designing these schemes but also identifying, correctly and transparently, which families should receive them.

Wrong identification causes two serious errors: inclusion errors (benefits reaching families that do not need them) and exclusion errors (genuinely needy families being left out). A rule-based knowledge system built on propositional logic offers a clear, auditable and consistent way to reduce both. Every decision can be traced back to a small set of true/false facts about the family and a set of explicit rules.

This assignment designs such a system. It defines the propositions (facts) that describe a household, constructs logical rules for exclusion, deprivation, overall eligibility and scheme-specific eligibility, and demonstrates the working of the system through truth tables and worked examples.

2. Objectives

- To represent the socio-economic condition of a family using atomic propositions.
- To build a knowledge base of logical rules that decides eligibility for poverty-alleviation schemes.
- To use inference (modus ponens and forward chaining) to derive eligibility conclusions automatically.
- To make decisions transparent, consistent and explainable, supporting SDG 1 and SDG 10 (Reduced Inequalities).

3. System Overview

The system follows the standard structure of a knowledge-based system, shown below.

Component	Role in this system
Fact base (input)	True/false values of the atomic propositions for one family, collected from a survey or application form.
Knowledge base (rules)	Propositional formulas that define exclusion, deprivation, eligibility and scheme-specific conditions.
Inference engine	Applies the rules to the facts using modus ponens and forward chaining until no new conclusions can be drawn.
Output	Eligible / not eligible / pending documentation, the list of schemes the family qualifies for, and the reasons.

Table 1: Components of the knowledge system

Scope and assumption. The criteria used here are a simplified, illustrative model inspired by the exclusion and deprivation criteria used in India's Socio-Economic and Caste Census (SECC 2011) and by schemes such as the Public Distribution System, PMAY housing, MGNREGA, social pension and health-insurance programmes. Actual thresholds and conditions differ by scheme and state and must be taken from official guidelines. In this model, numerical conditions (for example, income below the poverty line) are assumed to be evaluated beforehand and fed in as true/false facts.

4. Definition of Propositions

4.1 Atomic propositions (input facts)

Each atomic proposition describes one fact about a household and is either True (T) or False (F).

Symbol	Proposition (True when …)
RS	Family is a resident of the state / area covered by the scheme
LI	Household income is below the prescribed poverty threshold
GE	At least one member is a government employee
TX	At least one member pays income tax

Symbol	Proposition (True when …)
MV	Family owns a motorised four-wheeler or mechanised farm equipment
LD	Family owns agricultural land above the prescribed ceiling
KH	Family lives in a kutcha (temporary) house or is houseless
NE	There is no earning adult member in the age group 18–59
FH	Household is female-headed with no adult male member aged 18–59
DM	Household has a disabled member and no able-bodied adult
ML	Family is landless and earns its main income from casual manual labour
EM	Household has an elderly member (age 60 or above)
CH	Household has school-age children (6–14 years)
AW	An adult member is able and willing to do unskilled manual work
RU	Household is located in a rural area
AD	Family has valid identity documents (e.g. Aadhaar) and verified records
BK	Family has an active bank account
AB	Family is already availing the particular scheme being evaluated

Table 2: Atomic propositions

4.2 Derived propositions (conclusions)

Derived propositions are not entered directly; they are concluded by the rules in Section 5.

Symbol	Proposition (True when …)
EX	Family is automatically excluded (has a disqualifying economic status)
DEP	Family satisfies at least one deprivation indicator
ELG	Family is eligible for poverty-alleviation support (general eligibility)
VE	Family is eligible and its documents are verified (verified eligible)
PD	Family is eligible but documentation is pending
PRI	Family is in the priority (most vulnerable) category
CHK	Data inconsistency: record needs manual verification

Symbol	Proposition (True when …)
HOUS	Eligible for housing assistance
FOOD	Eligible for subsidised food grain / ration support
PENS	Eligible for social-security pension
EDUC	Eligible for education support (scholarship / free schooling aid)
EMPL	Eligible for guaranteed wage-employment scheme
DBT	Eligible to receive benefits through direct bank transfer

Table 3: Derived propositions

5. Logical Rules for Eligibility

The connectives used are: ¬ (NOT), ∧ (AND), ∨ (OR), → (implication) and ↔ (if and only if). The rule set is organised in layers: exclusion first, then deprivation, then general eligibility, and finally scheme-specific rules.

5.1 Exclusion rule

Rule	Formula	Meaning
R1	EX ↔ (GE ∨ TX ∨ MV ∨ LD)	A family is excluded if any one disqualifying condition is true: a government employee, an income-tax payer, ownership of a four-wheeler / mechanised equipment, or land above the ceiling.

5.2 Deprivation rule

Rule	Formula	Meaning
R2	DEP ↔ (KH ∨ NE ∨ FH ∨ DM ∨ ML)	A family shows deprivation if it lives in a kutcha house / is houseless, has no earning adult, is female-headed without an adult male, has a disabled member without an able-bodied adult, or is landless and dependent on manual labour.

5.3 General eligibility rules

Rule	Formula	Meaning
R3	ELG ↔ (RS ∧ ¬EX ∧ (LI ∨ DEP))	A resident family that is not excluded and is either low-income or deprived is eligible.
R4	VE ↔ (ELG ∧ AD)	An eligible family with valid, verified documents is verified-eligible.
R5	PD ↔ (ELG ∧ ¬AD)	An eligible family without valid documents is placed in the 'pending documentation' list.
R6	PRI ↔ (VE ∧ (NE ∨ FH ∨ DM))	Verified-eligible families with no earner, a female head or a disabled member get priority.
R7	CHK ↔ (LI ∧ (TX ∨ MV))	A family reported as low-income but also as a tax-payer or vehicle owner is contradictory; flag for manual verification.

5.4 Scheme-specific rules

Rule	Formula	Meaning
R8	HOUS ↔ (VE ∧ KH ∧ ¬AB)	Housing assistance for verified-eligible families living in kutcha houses or houseless, not already benefiting.
R9	FOOD ↔ (VE ∧ ¬AB)	Subsidised food grains for every verified-eligible family not already covered.
R10	PENS ↔ (VE ∧ EM ∧ ¬AB)	Social-security pension where an elderly member is present.
R11	EDUC ↔ (VE ∧ CH ∧ ¬AB)	Education support where school-age children are present.
R12	EMPL ↔ (VE ∧ RU ∧ AW ∧ ¬AB)	Guaranteed rural employment where an adult is able and willing to work.
R13	DBT ↔ (VE ∧ BK)	Benefits can be paid by direct bank transfer only if a bank account exists.

Tables 4–7: Rule base of the knowledge system

Note: AB is evaluated separately for each scheme (the family may already receive a pension but not housing). Rules R8–R12 are therefore applied once per scheme with the correct value of AB.

6. Truth Table for the Core Eligibility Rule

To verify the behaviour of Rule R3, let Q ≡ RS ∧ ¬EX (resident and not excluded). Then ELG ↔ Q ∧ (LI ∨ DEP). The truth table for all eight combinations is:

Q (resident & not excluded)	LI	DEP	ELG
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	F
F	T	F	F
F	F	T	F
F	F	F	F

Table 8: Truth table for ELG ↔ Q ∧ (LI ∨ DEP)

Observation: ELG is true only when Q is true and at least one of LI or DEP is true. Whenever Q is false (the family is a non-resident or is excluded), the family is ineligible regardless of income or deprivation. This confirms that the exclusion rule overrides all other conditions.

7. Inference Procedure

The system uses forward chaining, which begins from known facts and repeatedly fires rules whose conditions are satisfied until no new conclusions appear. The inference rules used are modus ponens (from P and P → Q, conclude Q) and the elimination of the biconditional (P ↔ Q gives P → Q and Q → P).

1. Read the truth values of all atomic propositions for the family (Table 2).
2. Apply R1 to decide EX; apply R2 to decide DEP.

3. Apply R3. If ELG is false, stop and report 'Not eligible' together with the reason (EX true, or RS false, or neither LI nor DEP true).
4. If ELG is true, apply R4 and R5 to decide between VE and PD. If PD is true, request the missing documents and stop.
5. If VE is true, apply R6 to find the priority status, and R8–R13 to list all schemes the family qualifies for.
6. Apply R7 to every record to flag contradictory data for manual verification.
7. Output the decision, the list of schemes and the chain of rules that produced it (explanation).

8. Worked Examples

Four sample families are evaluated. Only the propositions that are True are listed; all others are False.

Family	True facts	Key conclusions	Decision
A	RS, LI, KH, ML, CH, RU, AW, AD, BK	EX = F, DEP = T, ELG = T, VE = T	Eligible: HOUS, FOOD, EDUC, EMPL, DBT
B	RS, LI, NE, EM, AD, BK	EX = F, DEP = T, ELG = T, VE = T, PRI = T	Eligible (priority): FOOD, PENS, DBT
C	RS, LI, GE, AD, BK	EX = T, ELG = F	Not eligible: excluded because a member is a government employee
D	RS, DM, KH, CH, BK	EX = F, DEP = T, ELG = T, AD = F, VE = F, PD = T	Pending documentation; no scheme released until documents are verified

Table 9: Evaluation of sample families

Step-by-step derivation for Family A

1. Facts: RS, LI, KH, ML, CH, RU, AW, AD, BK are True; GE, TX, MV, LD, NE, FH, DM, EM, AB are False.
2. From R1: GE ∨ TX ∨ MV ∨ LD is False, so EX is False (¬EX is True).
3. From R2: KH is True, so KH ∨ NE ∨ FH ∨ DM ∨ ML is True, hence DEP is True.

4. From R3: RS ∧ ¬EX ∧ (LI ∨ DEP) = T ∧ T ∧ T, hence ELG is True.
5. From R4: ELG ∧ AD is True, hence VE is True.
6. From R8: VE ∧ KH ∧ ¬AB is True, so the family is eligible for housing assistance (HOUS).
7. From R9, R11, R12 and R13: FOOD, EDUC, EMPL and DBT are True. R10 fails because EM is False, so PENS is False.
8. From R6: none of NE, FH or DM is True, so PRI is False; the family is eligible but not in the priority category.

Why Family D is not rejected

Family D has a disabled member and no able-bodied adult (DM) and lives in a kutcha house (KH), so it is clearly deprived even though it is not reported as low-income (LI is False). Rule R3 therefore makes it eligible, and the missing identity documents only move it to the pending list (R5). This is how the system helps reduce exclusion errors: a vulnerable family is not silently dropped for lack of paperwork but is flagged for assistance in completing its documents.

9. Advantages and Limitations

Advantages

- Transparency: every decision can be explained by the rules that fired.
- Consistency: identical facts always lead to identical outcomes, reducing discretionary bias and corruption.
- Modularity: rules can be added or changed for a new scheme without rewriting the rest of the system.
- Reduces errors: the exclusion rule limits inclusion errors, while the deprivation rule and pending list limit exclusion errors.

Limitations

- Propositional logic cannot express quantities directly; income, land area or age thresholds must be converted into true/false facts beforehand.

- It treats a household as a single unit. Reasoning over individual members (for example, 'there exists a member aged 60 or above') needs first-order logic.
- Accuracy depends on the correctness of the survey data. Rule R7 helps detect contradictions but cannot replace field verification.
- Real schemes have many more conditions and state-specific variations; this model is a simplified prototype.

Possible extensions

The system can be extended using first-order logic for member-level reasoning, fuzzy logic or weighted scoring for borderline cases, and integration with digital databases (identity, bank and land records) to supply the facts automatically.

10. Conclusion

This assignment presented a propositional logic-based knowledge system for identifying families eligible for government poverty-alleviation schemes. Eighteen atomic propositions describe the household, thirteen derived propositions capture the conclusions, and a layered rule base of thirteen rules decides exclusion, deprivation, general eligibility, priority and scheme-specific entitlements. The truth table and worked examples show that the system behaves correctly, gives explainable decisions, and handles special cases such as excluded families and families with pending documents. Such a system supports SDG 1 by directing social protection to those who need it most, fairly and transparently.

References

1. United Nations. Transforming Our World: The 2030 Agenda for Sustainable Development – Goal 1: End poverty in all its forms everywhere.
2. Ministry of Rural Development, Government of India. Socio-Economic and Caste Census (SECC) 2011 – exclusion and deprivation criteria.
3. Government of India. National Food Security Act, 2013.
4. Russell, S. and Norvig, P. Artificial Intelligence: A Modern Approach (chapter on propositional logic and knowledge-based agents). Pearson.
5. Rosen, K. H. Discrete Mathematics and Its Applications (chapter on propositional logic). McGraw-Hill.

RUBRICS FOR CALCULATION:

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient